

# DEVOPS PROJECT

## Project Overview

To set up a complete CI/CD pipeline that automatically builds and deploys a static website from GitHub to production servers (Jenkins Master and a Node) using Jenkins, Docker, and GitHub Webhooks.

## EC2 INSTANCES

The screenshot displays the AWS Management Console for the 'eu-west-3' region. The 'Instances' page shows three running EC2 instances: 'slave-machine', 'Jenkins-Master', and 'Dev-Server', all of type 't2.micro'. Below the table, the 'slave-machine' instance is selected, showing its metrics: CPU utilization (approx. 13%), Network In (approx. 22.6M bytes), Network Out (approx. 579k bytes), and Network Packets In (approx. 15.6k). The console also shows a notification for a successfully deleted key pair and a 'Launch instances' button.

## GitHub Repository

The screenshot shows a GitHub repository named 'repo1' by user 'Snehal-sheke25'. The repository is public and has 4 commits. The commit history shows files being added: 'images', 'Dockerfile', 'index.html', and 'style.css'. The repository also has a 'README' section with a placeholder for adding a README file. The right sidebar shows the repository's activity, including 0 stars, 0 watching, and 0 forks. The repository is currently in the 'master' branch.

# Git Webhook

Black and White x Instances | EC2 x DevOps project x Webhooks - Setti x Dashboard - Jeni x Snehalshelke Swi x Sne Swim Acad x Cloud computin x +

github.com/Snehal-shelke25/repo1/settings/hooks

Apps All Bookmarks

Snehal-shelke25 / repo1

Type to search

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models Preview

Webhooks

Copilot

Environments

Codespaces

Pages

Security

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

✓ [http://13.39.14.188:8080/github-we...](#) (push) Edit Delete

Last delivery was successful.

26°C Mostly clear

vice president...

ENG IN

20:11 09-09-2025

# DEV Server

```
root@Dev-Server: ~/snehal-swim-academy
1 cd /root
2 wget "https://github.com/Snehal-shelke25/snehal-swim-site/raw/refs/heads/main/snehal-swim-academy.zip" -O website.zip
3 apt install unzip
4 unzip website.zip -d snehal-swim-academy
5 ls
6 cd snehal-swim-academy
7 ls
8 vim Dockerfile
9 apt install -y git
10 git init
11 git status
12 git add *
13 git status
14 git commit -m "adding all files"
15 git config --global user.name "Snehal Shelke"
16 git config --global user.email "snehalshelke@gmail.com"
17 git add *
18 git status
19 git commit -m "adding all files"
20 git status
21 ls
22 git remote add origin https://github.com/Snehal-shelke25/repo1.git
23 git push origin master
24 hostname Dev-Server
25 bash
26 cd /root
27 ls
28 vim index.html
29 cd snehal-swim-academy
30 ls
31 vim index.html
32 git status
33 git add *
34 git status
35 git commit -m "adding file"
36 git push origin master
37 vim index.html
38 git add *
39 git status
40 git commit -m "adding file"
41 git push origin master
42 vim index.html
43 cd snehal-swim-academy
44 ls
45 vim index.html
46 git add *
47 git status
48 git commit -m "adding file"
49 git status
50 git push origin master
51 vim index.html
52 git status
53 git add *
54 git status
55 git commit -m "adding file"
56 git status
57 git push origin master
58 history'
```

26°C Mostly clear

vice president...

ENG IN

20:03 09-09-2025

# Jenkins Master

```
root@Jenkins-Master:~# history
1 apt update -y
2 apt upgrade -y
3 curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null
4 echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
5 apt install openjdk-17-jdk -y
6 apt install jenkins -y
7 history
8 apt install jenkins -y
9 sudo apt update -y
10 sudo apt upgrade -y
11 sudo apt install -y fontconfig openjdk-11-jre
12 curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null
13 echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
14 sudo apt update -y
15 sudo apt install -y jenkins
16 systemctl status jenkins
17 apt list jenkins
18 systemctl enable jenkins
19 systemctl start jenkins
20 apt install git -y
21 apt install docker.io -y
22 systemctl start docker
23 systemctl enable docker
24 history
25 vim /etc/sudoers
26 cat /var/lib/jenkins/secrets/initialAdminPassword
27 hostname Jenkins-Master
28 bash
29 history
root@Jenkins-Master:~#
```

# Slave machine

```
root@slave-machine:/dir1# history
1 hostname slave-machine
2 bash
3 apt update -y
4 apt upgrade -y
5 apt install openjdk-17-jdk -y
6 apt install git -y
7 apt install docker.io -y
8 systemctl start docker
9 systemctl enable docker
10 history
11 mkdir /dir1
12 cd /dir1
13 curl -sO http://13.39.14.188:8080/jnlpJars/agent.jar
14 java -jar agent.jar -url http://13.39.14.188:8080/ -secret 4dd741f61ba3c7453e4d2299494619b16fc4252f3f8ecfbbbd5911aa2bfce4b5 -name slave -webSocket -workDir "/ dir1"
15 nohup java -jar agent.jar -jnlpUrl http://13.39.14.188:8080/computer/slave/slave-agent.jnlp -secret abcd1234efgh5678 -workDir "/home/ubuntu/agent" > agent.log 2>&1 &
16 nohup curl -sO http://13.39.14.188:8080/jnlpJars/agent.jar
17 java -jar agent.jar -url http://13.39.14.188:8080/ -secret 4dd741f61ba3c7453e4d2299494619b16fc4252f3f8ecfbbbd5911aa2bfce4b5 -name slave -webSocket -workDir "/ dir1" &
18 history
root@slave-machine:/dir1#
```

# Freestyle Project for Built-in server (Website with Job status)

The image displays two screenshots of a web browser interface. The top screenshot shows the homepage of the Snehal Swim Academy. The page features a background image of a swimmer in a pool. At the top center, there is a logo for 'SNEHAL SWIM ACADEMY'. Below the logo, the text 'Welcome to Snehal Swim Academy' is displayed, followed by the tagline 'Learn to swim with confidence and fun!'. A central section titled 'Our Services' lists four items: Kids Classes, Adults Training, Competition Coaching, and Aquatic Fitness. Below this, a 'Contact Us' section provides an email address (snesheike@gmail.com) and a phone number (8788084375). The bottom screenshot shows the Jenkins web interface for a job named 'job1'. The left sidebar contains navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, GitHub Hook Log, and Rename. The main content area shows the job status as 'job1' with a green checkmark. Below this, there is a 'Permalinks' section with four links: 'Last build (#1), 50 sec ago', 'Last stable build (#1), 50 sec ago', 'Last successful build (#1), 50 sec ago', and 'Last completed build (#1), 50 sec ago'. A 'Builds' section is also visible, showing a list of builds with a filter input. The bottom status bar of the browser shows the temperature as 26°C and the time as 18:16 on 09-09-2025.

**Snehal Swim Academy**

Welcome to Snehal Swim Academy

Learn to swim with confidence and fun!

**Our Services**

- Kids Classes
- Adults Training
- Competition Coaching
- Aquatic Fitness

**Contact Us**

Email: snesheike@gmail.com

Phone: 8788084375

**Jenkins / job1**

Status **job1** Add description

Changes

Workspace

Build Now

Configure

Delete Project

GitHub Hook Log

Rename

**Permalinks**

- Last build (#1), 50 sec ago
- Last stable build (#1), 50 sec ago
- Last successful build (#1), 50 sec ago
- Last completed build (#1), 50 sec ago

**Builds**

Filter

Today

- #2 12:46 PM
- #1 12:37 PM

REST API Jenkins 2.516.2

13.39.14.188:8080/job/job1/#

26°C Sunny

vice president...

18:16 09-09-2025

# Modify Index.html file

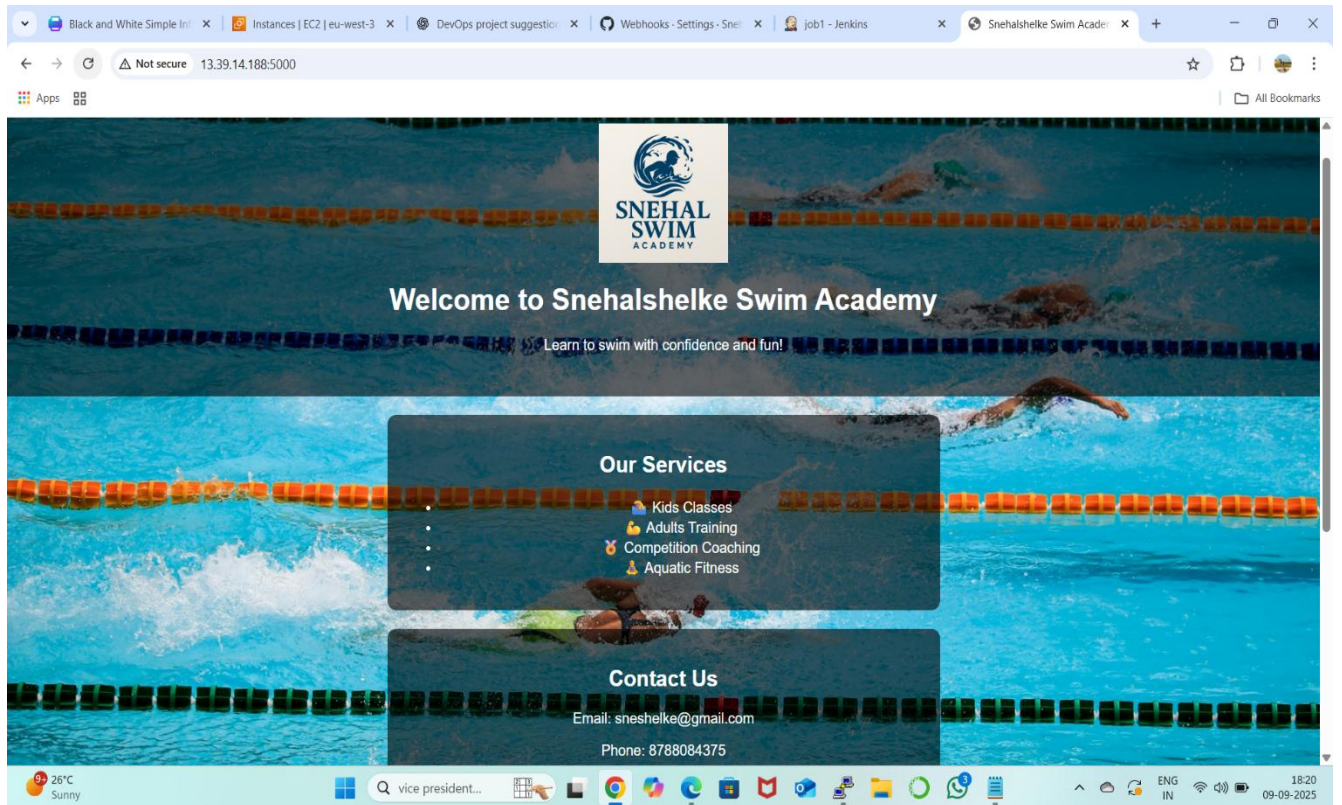
```
root@Dev-Server: ~/snehal-swim-academy
1 file changed, 2 insertions(+), 2 deletions(-)
root@Dev-Server:~/snehal-swim-academy# git status
On branch master
nothing to commit, working tree clean
root@Dev-Server:~/snehal-swim-academy# git push origin master
Username for 'https://github.com': Snehal-shelke25
Password for 'https://Snehal-shelke25@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 291 bytes | 291.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Snehal-shelke25/repol.git
4badc49..97df197 master -> master
root@Dev-Server:~/snehal-swim-academy# vim index.html
root@Dev-Server:~/snehal-swim-academy# git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
root@Dev-Server:~/snehal-swim-academy# git add *
root@Dev-Server:~/snehal-swim-academy# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   index.html

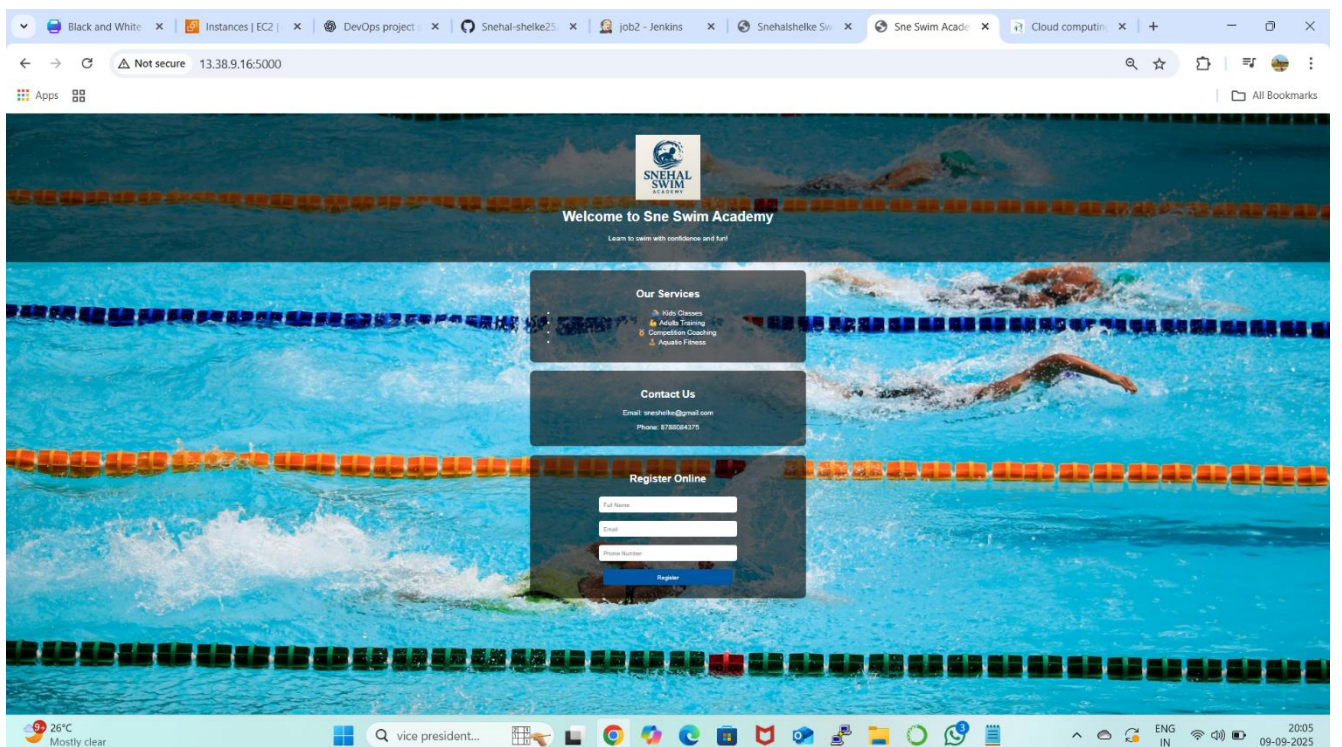
root@Dev-Server:~/snehal-swim-academy# git commit -m "adding file"
[master 462abd3] adding file
1 file changed, 2 insertions(+), 2 deletions(-)
root@Dev-Server:~/snehal-swim-academy# git status
On branch master
nothing to commit, working tree clean
root@Dev-Server:~/snehal-swim-academy# git push origin master
Username for 'https://github.com': Snehal-shelke25
Password for 'https://%20Snehal-shelke25@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 310 bytes | 310.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Snehal-shelke25/repol.git
97df197..462abd3 master -> master
root@Dev-Server:~/snehal-swim-academy#
```



# Freestyle Project for Node server (Website with Job status)



# Multi-configuration project for Built-in & Node server (Job status with Website)



The image is a screenshot of a web browser displaying the Jenkins CI/CD interface. The browser's address bar shows the URL 'https://13.39.14.188:8080/job/job2/'. The page title is 'job2 - Jenkins'. The Jenkins logo is visible in the top left corner. The main content area is titled 'job2' and shows a green checkmark icon. Below this, there are two tabs: 'built-in' and 'slave', both with green checkmarks. A section titled 'Permalinks' lists four links: 'Last build (#2), 5 min 44 sec ago', 'Last stable build (#2), 5 min 44 sec ago', 'Last successful build (#2), 5 min 44 sec ago', and 'Last completed build (#2), 5 min 44 sec ago'. On the left sidebar, there are several navigation options: 'Status', 'Changes', 'Workspace', 'Build Now', 'Configure', and 'Delete Multi-configuration project'. At the bottom of the sidebar, there is a 'Builds' section with a search filter and a list of builds for 'Today'. The list shows two builds: '#2' at '2:29 PM' and '#1' at '2:28 PM'. The bottom of the image shows a Windows taskbar with various icons, including the Start button, search bar, and several application icons. The system tray shows the temperature as '26°C Mostly clear' and the date/time as '20:06 09-09-2025'.

**THANK YOU**